

Within-firm exhibits — Tables A6, A7 and A8

Stata v2 port. Estimated by `01b_within_firm_estimates{, _hw}.do` in `07_within_firm/` of the canonical pipeline, from the canonical inputs. The LaTeX is produced by `05b_make_tables_within_firm.py`, unmodified, over the CSVs in `Tables/layer_connectivity/07_within_firm/`. The pre-revision estimates are archived under `archive_oldspec_2026-07-31/` in that same directory.

Caveat. The table notes below are inherited unchanged from `05b_make_tables_within_firm.py` and still describe the *old* fixed-effect list. Under the revised specification the industry, microregion and negotiation-month year interactions are group-interacted, and the “Overall” column additionally carries the firm-level pre-treatment bins. The numbers are correct; this prose is not, and must be rewritten before the tables go into the paper.

Table A7 revision. Three controls are added to the two group-level columns; nothing is removed. See `SPEC.md` for the exact fixed-effect lists.

1. The year interactions are allowed to differ by worker group: $\text{group} \times \text{industry} \times \text{year}$, $\text{group} \times \text{microregion} \times \text{year}$, $\text{group} \times \text{negotiation-month} \times \text{year}$. The plain versions are then omitted, being nested inside these.
2. The three pre-treatment quartile bins are made group-specific. They are already computed per $\text{establishment} \times \text{group}$; only their year path was shared.
3. The “Overall” column gains the firm-level pre-treatment wage, employment and flow bins $\times \text{year}$, the same ones Table A7 column (1) uses. “Within firms” absorbs them through $\text{establishment} \times \text{year}$.

Unchanged. Table A6, the firm-level columns of Table A7, and Table A8 are firm-level objects and are not affected.

Table A6 — Group-Level Connectivity Descriptives (monthly wages)

Group-Level Connectivity Descriptives

	<i>Education</i>		<i>Gender</i>		<i>Tenure</i>	
	No high school	High school+	Female	Male	<12 months	≥12 months
Avg. employment	60.2	104.3	55.7	107.9	48.8	110.1
Avg. wage	1,796	2,613	2,149	2,596	2,054	2,610
Avg. per-worker flows	0.2055	0.2732	0.2449	0.2687	0.5208	0.1083
Connectivity	0.0094	0.0148	0.0103	0.0149	0.0259	0.0067
% firms w/ both groups	85.8%		88.0%		96.5%	
% of connectivity variance within firms	27.5%		33.8%		45.3%	
% of connectivity variance between firms	72.5%		66.2%		54.7%	

Notes: This table summarizes pre-treatment (2009–2011) descriptive statistics for group-level connectivity under three worker partitions. The unit of observation is a group $\times$ firm cell. Tenure groups are defined by whether a worker has completed twelve months at the establishment. “Avg. per-worker flows” is total worker flows per group worker, averaged across year pairs 2007–08 through 2010–11. “Connectivity” is the average pre-treatment worker flow between a firm’s group- g workers and directly treated establishments, per group- g worker; a firm’s connectivity therefore varies across its worker groups. “% firms w/ both groups” is the share of spillover-sample firms with non-missing connectivity for both groups. The final two rows decompose the total variance of group connectivity into its within- and between-firm components on firms with both groups, following the ANOVA identity $\text{Var}(X) = \text{E}[\text{Var}(X \mid \text{firm})] + \text{Var}(\text{E}[X \mid \text{firm}])$, and report each as a share of the total.

Table A7 — Group-level spillover effects, revised specification (log monthly wages)

Group-level spillover effects — education, gender, tenure

	Log Wages			Log Employment		
	Firm-level (1)	Within firms (2)	Overall (3)	Firm-level (4)	Within firms (5)	Overall (6)
Panel A: Education (no HS / HS+)						
Connectivity × Post	0.0050** (0.0022)			0.0008 (0.0081)		
Group-Connectivity × Post		−0.0032 (0.0044)	0.0028 (0.0023)		−0.0123 (0.0102)	−0.0070 (0.0068)
Observations	32,498	50,612	57,789	32,704	50,612	57,789
Groups × firms	—	6,938	7,565	—	6,938	7,565
Firms	4,085	3,469	4,085	4,088	3,469	4,085
Pre-trend (placebo)	0.0019 (0.0025)	0.0048 (0.0057)	0.0017 (0.0028)	−0.0015 (0.0063)	0.0101 (0.0101)	0.0041 (0.0051)
Panel B: Gender (female / male)						
Connectivity × Post	0.0050** (0.0022)			0.0008 (0.0081)		
Group-Connectivity × Post		0.0020 (0.0025)	0.0028 (0.0018)		−0.0006 (0.0047)	0.0038 (0.0048)
Observations	32,498	53,592	59,289	32,704	53,592	59,289
Groups × firms	—	7,156	7,672	—	7,156	7,672
Firms	4,085	3,578	4,085	4,088	3,578	4,085
Pre-trend (placebo)	0.0019 (0.0025)	−0.0010 (0.0054)	−0.0002 (0.0024)	−0.0015 (0.0063)	−0.0067 (0.0110)	−0.0022 (0.0061)
Panel C: Tenure (<12mo / ≥12mo)						
Connectivity × Post	0.0050** (0.0022)			0.0008 (0.0081)		
Group-Connectivity × Post		0.0013 (0.0025)	0.0017 (0.0021)		−0.0038 (0.0067)	0.0025 (0.0059)
Observations	32,498	54,650	59,786	32,704	54,650	59,786
Groups × firms	—	7,846	8,042	—	7,846	8,042
Firms	4,085	3,923	4,085	4,088	3,923	4,085
Pre-trend (placebo)	0.0019 (0.0025)	0.0017 (0.0039)	0.0038 (0.0026)	−0.0015 (0.0063)	−0.0108 (0.0089)	−0.0106 (0.0065)
Additional Controls						
Group-level Variables		✓	✓		✓	✓
Firm × Year FE		✓			✓	

Notes: Each panel partitions workers into two groups—by education (Panel A), gender (Panel B), or tenure (Panel C)—and estimates spillover effects at the firm and group level, on the sample of untreated, balanced-panel establishments. Group-Connectivity is defined analogously to firm-level Connectivity, replacing total firm employment with group-specific employment when normalizing worker flows to directly treated establishments. Both measures are scaled so that a value of 1 corresponds to the 90th percentile of the firm-level distribution. In the “Within firms” and “Overall” columns, the unit of observation is a group × firm × year cell and all regressions include group × firm fixed effects. “Within firms” columns add firm × year fixed effects; “Overall” columns instead include microregion × year, industry × year, and negotiation-month × year fixed effects. Group-level controls are year-interacted quartile bins of pre-treatment group wage, group employment, and group per-worker flows. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). Standard errors clustered at the establishment level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table A8 — Group-Specific Connectivity and Firm-Level Outcomes (log monthly wages)

Group-Specific Connectivity and Firm-Level Outcomes				
<i>Firm-level outcome:</i>	Post × Connectivity		Pre-trend (placebo)	
	Log Wages (1)	Log Employment (2)	Log Wages (3)	Log Employment (4)
Panel A: Education (no HS / HS+)				
Low-Education Connectivity	0.0027 (0.0018)	0.0021 (0.0076)	−0.0010 (0.0025)	0.0010 (0.0067)
High-Education Connectivity	0.0020 (0.0022)	0.0035 (0.0076)	0.0020 (0.0023)	0.0025 (0.0056)
Equality test (<i>p</i> -value)	0.839	0.901		
Observations	28,227	28,368		
Firms	3,546	3,546		
Panel B: Gender (female / male)				
Female Connectivity	0.0012 (0.0011)	0.0025 (0.0031)	0.0006 (0.0016)	0.0011 (0.0025)
Male Connectivity	0.0020 (0.0016)	0.0018 (0.0051)	0.0020 (0.0028)	−0.0008 (0.0057)
Equality test (<i>p</i> -value)	0.712	0.911		
Observations	28,903	29,056		
Firms	3,632	3,632		
Panel C: Tenure (<12mo / ≥12mo)				
Low-Tenure Connectivity	0.0006 (0.0013)	−0.0034 (0.0039)	0.0021 (0.0013)	−0.0006 (0.0034)
High-Tenure Connectivity	0.0052** (0.0022)	0.0090 (0.0085)	0.0002 (0.0024)	0.0034 (0.0056)
Equality test (<i>p</i> -value)	0.078	0.178		
Observations	31,829	32,000		
Firms	4,000	4,000		

Notes: Each panel partitions workers into two groups and jointly estimates the spillover effects of the two resulting group-specific connectivity measures on firm-level outcomes, on the sample of untreated establishments. Low-education workers are those without a high school degree and high-education workers those with at least a high school degree; low-tenure workers are those with fewer than twelve months at the establishment and high-tenure workers those with twelve months or more. Both measures enter regressions simultaneously. Group-specific connectivity is defined as in the group-level spillover table, and both measures are scaled so that a value of 1 corresponds to the 90th percentile of the firm-level distribution. The unit of observation is a firm × year cell, and all regressions include establishment fixed effects, year-interacted pre-treatment quartile bins of firm wage, firm employment and per-worker flows, and industry, microregion and negotiation-month interactions with year. Columns (1)–(2) report average effects for 2012–2016, with 2011 as the reference year. Columns (3)–(4) report coefficients from placebo regressions estimated on pre-treatment data only (2009–2010 relative to 2011); the equality test, observation counts, and firm counts refer to the specifications in columns (1)–(2). Equality test reports the *p*-value of the test that the two coefficients in the column are equal, using the normal approximation. Samples are smaller than in the main spillover table because both group-specific measures must be non-missing. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6 — Group-Level Connectivity Descriptives (hourly wages)

Group-Level Connectivity Descriptives — hourly wages

	<i>Education</i>		<i>Gender</i>		<i>Tenure</i>	
	No high school	High school+	Female	Male	<12 months	≥12 months
Avg. employment	60.2	104.3	55.7	107.9	48.8	110.1
Avg. hourly wage	9.9	14.8	12.2	14.7	11.7	14.7
Avg. per-worker flows	0.2055	0.2732	0.2449	0.2687	0.5208	0.1083
Connectivity	0.0094	0.0148	0.0103	0.0149	0.0259	0.0067
% firms w/ both groups	85.8%		88.0%		96.5%	
% of connectivity variance within firms	27.5%		33.8%		45.3%	
% of connectivity variance between firms	72.5%		66.2%		54.7%	

Notes: This table summarizes pre-treatment (2009–2011) descriptive statistics for group-level connectivity under three worker partitions. The unit of observation is a group $\times$ firm cell. The average hourly wage row reports the level implied by the corrected layer-level hourly wage outcome, computed from natural-log real hourly wages using monthly hours equal to `horascontr` $\times$ 4.348. Tenure groups are defined by whether a worker has completed twelve months at the establishment. “Avg. per-worker flows” is total worker flows per group worker, averaged across year pairs 2007–08 through 2010–11. “Connectivity” is the average pre-treatment worker flow between a firm’s group- g workers and directly treated establishments, per group- g worker; a firm’s connectivity therefore varies across its worker groups. “% firms w/ both groups” is the share of spillover-sample firms with non-missing connectivity for both groups. The final two rows decompose the total variance of group connectivity into its within- and between-firm components on firms with both groups, following the ANOVA identity $\text{Var}(X) = \text{E}[\text{Var}(X \mid \text{firm})] + \text{Var}(\text{E}[X \mid \text{firm}])$, and report each as a share of the total.

Table A7 — Group-level spillover effects, revised specification (log hourly wages)

Group-level spillover effects — education, gender, tenure (log hourly wages)						
	Log Hourly Wages			Log Employment		
	Firm-level (1)	Within firms (2)	Overall (3)	Firm-level (4)	Within firms (5)	Overall (6)
Panel A: Education (no HS / HS+)						
Connectivity × Post	0.0065*** (0.0023)			0.0008 (0.0081)		
Group-Connectivity × Post		−0.0039 (0.0043)	0.0024 (0.0023)		−0.0123 (0.0102)	−0.0070 (0.0068)
Observations	32,498	50,612	57,789	32,704	50,612	57,789
Groups × firms	—	6,938	7,565	—	6,938	7,565
Firms	4,085	3,469	4,085	4,088	3,469	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0054 (0.0057)	0.0012 (0.0029)	−0.0015 (0.0063)	0.0101 (0.0101)	0.0041 (0.0051)
Panel B: Gender (female / male)						
Connectivity × Post	0.0065*** (0.0023)			0.0008 (0.0081)		
Group-Connectivity × Post		0.0023 (0.0022)	0.0046** (0.0019)		−0.0006 (0.0047)	0.0038 (0.0048)
Observations	32,498	53,592	59,289	32,704	53,592	59,289
Groups × firms	—	7,156	7,672	—	7,156	7,672
Firms	4,085	3,578	4,085	4,088	3,578	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0002 (0.0053)	0.0002 (0.0024)	−0.0015 (0.0063)	−0.0067 (0.0110)	−0.0022 (0.0061)
Panel C: Tenure (<12mo / ≥12mo)						
Connectivity × Post	0.0065*** (0.0023)			0.0008 (0.0081)		
Group-Connectivity × Post		−0.0007 (0.0024)	0.0024 (0.0023)		−0.0038 (0.0067)	0.0025 (0.0059)
Observations	32,498	54,650	59,786	32,704	54,650	59,786
Groups × firms	—	7,846	8,042	—	7,846	8,042
Firms	4,085	3,923	4,085	4,088	3,923	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0036 (0.0038)	0.0048* (0.0026)	−0.0015 (0.0063)	−0.0108 (0.0089)	−0.0106 (0.0065)
Additional Controls						
Group-level Variables		✓	✓		✓	✓
Firm × Year FE		✓			✓	

Notes: Hourly-wage counterpart of the group-level spillover table. The firm-level hourly spillover benchmark is 0.0065 with standard error 0.0023. The “Within firms” columns add firm × year fixed effects, so identification comes from differential group connectivity across groups within the same firm-year; the “Overall” columns instead use industry, microregion, and negotiation-month interactions with year. Group-level employment columns are reported for completeness but remain mechanically division-biased because group employment enters the connectivity denominator. Standard errors are clustered by establishment. Hourly wages are natural-log real hourly wages, computed with monthly hours equal to `horascontr` × 4.348.

Table A8 — Group-Specific Connectivity and Firm-Level Outcomes (log hourly wages)

Group-Specific Connectivity and Firm-Level Outcomes — Log Hourly Wages

<i>Firm-level outcome:</i>	Post × Connectivity		Pre-trend (placebo)	
	Log Hourly Wages (1)	Log Employment (2)	Log Hourly Wages (3)	Log Employment (4)
Panel A: Education (no HS / HS+)				
Low-Education Connectivity	0.0027 (0.0021)	0.0021 (0.0076)	−0.0008 (0.0026)	0.0010 (0.0067)
High-Education Connectivity	0.0033 (0.0024)	0.0035 (0.0076)	0.0017 (0.0024)	0.0025 (0.0056)
Equality test (<i>p</i> -value)	0.875	0.901		
Observations	28,227	28,368		
Firms	3,546	3,546		
Panel B: Gender (female / male)				
Female Connectivity	0.0019* (0.0011)	0.0025 (0.0031)	0.0006 (0.0016)	0.0011 (0.0025)
Male Connectivity	0.0027 (0.0017)	0.0018 (0.0051)	0.0017 (0.0027)	−0.0008 (0.0057)
Equality test (<i>p</i> -value)	0.720	0.911		
Observations	28,903	29,056		
Firms	3,632	3,632		
Panel C: Tenure (<12mo / ≥12mo)				
Low-Tenure Connectivity	0.0005 (0.0013)	−0.0034 (0.0039)	0.0021 (0.0013)	−0.0006 (0.0034)
High-Tenure Connectivity	0.0088*** (0.0028)	0.0090 (0.0085)	0.0002 (0.0024)	0.0034 (0.0056)
Equality test (<i>p</i> -value)	0.010	0.178		
Observations	31,829	32,000		
Firms	4,000	4,000		

Notes: Hourly-wage counterpart of the horse-race table. Each panel partitions workers into two groups and jointly estimates the spillover effects of the two resulting group-specific connectivity measures on firm-level outcomes, on the sample of untreated establishments. Low-education workers are those without a high school degree and high-education workers those with at least a high school degree; low-tenure workers are those with fewer than twelve months at the establishment and high-tenure workers those with twelve months or more. Both measures enter regressions simultaneously. Group-specific connectivity is defined as in the group-level spillover table, and both measures are scaled so that a value of 1 corresponds to the 90th percentile of the firm-level distribution. The unit of observation is a firm × year cell, and all regressions include establishment fixed effects, year-interacted pre-treatment quartile bins of firm wage, firm employment and per-worker flows, and industry, microregion and negotiation-month interactions with year. Columns (1)–(2) report average effects for 2012–2016, with 2011 as the reference year. Columns (3)–(4) report coefficients from placebo regressions estimated on pre-treatment data only (2009–2010 relative to 2011); the equality test, observation counts, and firm counts refer to the specifications in columns (1)–(2). Equality test reports the *p*-value of the test that the two coefficients in the column are equal, using the normal approximation. Samples are smaller than in the main spillover table because both group-specific measures must be non-missing. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.